DiaTwin: AI-Powered Insulin Dosing

Profile for EXIST Open Consultation

1. Team & Skills

Contact Details

Field	Founder 1 – CEO & CPO	Co-Founder 2 – CMO
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Address	Dorfen , Greater Munich Area	Greater Munich Area
Education	M.Sc. Digital Health (DIT, grade 1.6) Health Informatics Diploma Bachelor Pharmacy (Zagazig University)	M.Sc. Digital Health (DIT, final semester, graduating June 2026) Doctor of Medicine, Arba Minch University (Distinction)
Personal situation	Egyptian national, job search visa valid until Feb 2027	Ethiopian national, based in Munich; final semester student

Is Your Team Complete?

Yes. The two-person team covers all core competencies required for DiaTwin:

Competency	Covered by
Product development & AI algorithms	Ali Attia (pharmacist, Digital Health M.Sc.)
Diabetes & insulin therapy expertise	Ali Attia (pharmacist specializing in insulin therapy)
Clinical validation & medical oversight	Saron Gelashet (MD, Digital Health M.Sc.)
Regulatory & GCP compliance	Saron Gelashet (ICH GCP E6(R3) certified)
MedTech & data quality	Saron Gelashet (medical AI annotation, Snke OS GmbH)

The one remaining gap , deep knowledge of the German healthcare system for DiGA certification and insurance negotiations will be addressed through our mentors , and the coaching and training provided through the EXIST program during the funding year. We have already begun familiarizing ourselves with the current DiGA regulatory pathway and requirements.

2. Motivation & Personal Drive

Why pursue this startup?

Multiple family members have Type 1 Diabetes. Watching them struggle daily with insulin dosing, and knowing that precision tools cost €10,000–16,000 in Egypt, making them completely unaffordable , made this personal.

As a pharmacist specializing in insulin therapy, I realized a smartphone could deliver the same precision as an insulin pump using just cheap test strips. I built DiaTwin during my master's thesis, validated by Prof. Dr. Klaus Parhofer (LMU Munich Chief of Endocrinology), who rated it 8.7/10 for clinical utility and graded the thesis "Excellent."

Mission: Make precision diabetes care accessible to everyone — reducing complications and enabling data-driven decisions for doctors.

Why now?

1. **Regulatory opportunity:** Germany's DiGA framework enables €1,800/year insurance reimbursement per patient, sustainable revenue without out-of-pocket costs.
2. **Market gap:** 340,000 German patients have discontinued expensive glucose sensors but have no precision alternative. No solution currently addresses this gap.
3. **Technical readiness:** MVP is complete and validated.
4. **First-mover advantage:** No competitor targets this patient group yet.

3. Startup Idea

What problem are you solving?

Insulin-dependent patients inject 4–6 times daily, but no affordable tool tells them how much insulin to take. Standard doctor rules ignore critical factors: insulin still active in the body, exercise effects, meal type, and time of day. The result is dangerous guesswork:

- Too little insulin → High blood sugar → Blindness, kidney failure, amputations
- Too much insulin → Low blood sugar → Confusion, seizures, coma

Target Group

- 340,000 post-sensor users (Germany): discontinued expensive CGMs (€3,000–5,000/year), no precision alternative
- 60,000 current sensor users: sensors show glucose trends but don't calculate doses
- 900,000 insulin pen users: Type 2 patients overwhelmed by complex calculations
- 5,000 endocrinologists (secondary): spend 30–60 min/patient reviewing paper logs

Solution: DiaTwin

DiaTwin is a smartphone app that calculates the exact insulin dose for each meal, with doctor supervision built in. The doctor configures personal insulin settings, and the patient logs blood sugar and activity, or syncs an existing glucose device and smart watch. The patient then photographs their meal or enters a simple description, and the app recommends the exact dose and timing (e.g. "Take 8 units now, 4 units in 90 minutes").

Key features:

- Works with ANY glucose device (finger-prick, Abbott, Dexcom, Medtronic, Accu-Chek)
- Tracks active insulin to prevent overdose
- Learns personal patterns (exercise effects, meal responses)
- Predicts blood sugar 4 hours ahead
- Doctor dashboard: 5-minute visit review vs. 30–60 minutes with paper logs

Cost: €120/year (finger-prick strips) vs. €3,000–5,000/year (sensors), same precision, 96% cheaper.

Business model: German health insurance pays €1,800/year per patient (€0 for patients). This revenue funds 100% free access in Egypt, India, and Kenya.

4. Current Status

☑Completed	Functional MVP (web + mobile apps) Core algorithms tested with 50,000+ meal calculations Master's thesis validation: Grade 1.6 "Excellent" from Prof. Dr. Klaus Parhofer (LMU Munich) Academic partnerships secured: Charité Berlin & LMU Munich Team complete (CEO/CPO + CMO)
In Progress	EXIST application preparation MVP refinement (doctor dashboard, data upload features) Familiarizing with DiGA regulatory pathway and requirements
⧖EXIST Year Plan	Clinical pilot study: 20 patients, 3 months CE marking + DiGA certification (18–24 months) 50–100 beta testers GmbH registration

5. Market & Competitors

Market Size

- Germany: 1.3 million insulin users; primary focus: 340,000 post-sensor patients. Year 5 target: 2,000 users × €1,800 DiGA = €3.6M/year
- Europe: 7 million insulin users; similar reimbursement frameworks in France, UK, Netherlands
- Global: 50+ million insulin users in developing countries (free tier)
- Industry: Global diabetes device market $35B (+8.7%/year); digital diabetes management $5B (+12%/year)

Competitive Landscape

Solution	Cost	Key Limitation
Insulin pumps (Medtronic, Tandem)	€10,000–16,000	Unaffordable for most patients
Sensors + apps (Abbott, Dexcom)	€3,000–5,000/yr	No dose calculation; 85% discontinue due to cost
Logbook apps (MySugr)	Free–€10/mo	No dose guidance, just data logging
Simple calculators	Free	No insulin-on-board tracking; no doctor oversight
Accu-Chek SmartGuide (main competitor)	€1,359/yr	Locked to Accu-Chek devices only; no physician dashboard

DiaTwin's Competitive Advantage

- Platform-agnostic: works with any existing glucose device — no lock-in, no extra hardware purchase
- Only solution for the 340,000-person post-sensor gap in Germany
- Physician workflow integration: saves 25 min/visit, so doctors recommend it
- DiGA pathway: €0 patient cost via insurance vs. €1,359/year out-of-pocket (SmartGuide)
- Global equity: German revenue funds free access in Egypt, India, Kenya

6. Expectations & Key Questions

What we'd like to discuss

- Application strength: is our current status strong enough for EXIST, or should we strengthen further before applying?
- Budget validation: is our allocation realistic? (€15K pilot study, €4K regulatory, €6.5K marketing)
- Next steps: what does the application process look like from here, and what does Johanna need from us?

Key Questions

- What is the expected timeline of the grant, from application submission to funding decision and first payment?
- Can co-founders be added or adjusted during EXIST Year 1?
- What do most successful EXIST alumni do after the funding year, bootstrap with revenue, raise VC funding, or get acquired?
- Academic affiliation: does EXIST provide access to university labs and academic mentors?

Consent:

I agree to be contacted by email, to confidential data processing for consulting purposes, and to photos/videos during events for documentation and PR purposes.